CampusFlow LMS

Storage deployment and easy upgrade guide

Short answer: the system has 3 core services. It does not currently prove that a separate file-storage server is attached. File uploads use the backend uploads folder by default, while Firebase Storage is an optional cloud mode.

Frontend service

1 Railway service Website files only

Backend service

1 Railway service API and uploads

Database service

1 PostgreSQL service Data only

1. How many storage locations are there?

Location	Count	Purpose	Current status
PostgreSQL	1	Users, courses, enrollments and payment records	Required and online
Backend /app/uploads	0 or 1	Course files, profiles, homework and other uploads	Code supports local storage; Railway volume not confirmed
Firebase Storage bucket	0 or 1	Cloud media for community posts, stories, chat and groups	Optional; depends on Railway variables
Telegram archive	0 or 1	Backup/archive only; not used to serve files	Optional

2. Current storage behavior

- The application writes many uploads to the backend folder: /app/uploads.
- Community media can use Firebase Storage when MEDIA_STORAGE is set to firebase or google.
- The exact Railway volume count and the production MEDIA_STORAGE value are external settings, so they cannot be proven from source code alone.
- Local workspace test only: backend/uploads contains 19 files, about 7.7 MB. This is not the production usage number.
- The live frontend and backend were last checked successfully with HTTP 200.

3. Simple picture

Users	Frontend	Backend	Storage
Browser or mobile	React website	NestJS API	PostgreSQL + uploads volume or Firebase bucket

4. Easiest upgrade now: add one Railway Volume

This is the lowest-code change because the current upload paths already use /app/uploads.

- Open Railway and select the Backend service.
- Choose Volumes -> Add Volume.
- Set the mount path to /app/uploads.
- Deploy the backend and test one profile image, one course file and one community post.
- If the service reports a permission error, set RAILWAY_RUN_UID=0 and redeploy.
- Do not delete the old files until the new volume is verified and backed up.

Railway storage sizes

Plan	Default volume size	Upgrade note
Free / Trial	0.5 GB	Small testing only
Hobby	5 GB	Good starting point for a small demo
Pro	Up to 1 TB self-service	Best for larger production media

Railway says paid-plan volumes can be resized live. A volume is attached to a service, so attach it to the Backend service, not the Frontend service.

5. Best long-term upgrade: Firebase Storage

Use one Firebase Storage bucket for scalable cloud media. The code already supports the cloud mode for community media.

- Enable Storage in the Firebase project.
- Set MEDIA_STORAGE=firebase in Railway backend variables.
- Set FIREBASE_PROJECT_ID and FIREBASE_STORAGE_BUCKET.
- Keep FIREBASE_CLIENT_EMAIL and FIREBASE_PRIVATE_KEY only in Railway Variables; never put them in the frontend or report.
- Upload a test story or post and confirm its URL uses firebasestorage.googleapis.com.
- Later migrate course files, homework and profile uploads so all user media uses the same cloud bucket.

6. My recommendation

Now	Add one Railway Volume at /app/uploads. This protects current uploads with minimal code change.
Later	Move all user media to one Firebase Storage bucket for easier scaling and multiple server replicas.

Official references

https://docs.railway.com/volumes/reference
https://docs.railway.com/volumes
https://firebase.google.com/docs/storage/web/start